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Why Zero-Emission Isn't Enough: A Fundamental Impact Factor for Urban Logistics

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System-Wide Health

How come natural transport systems (mineral transport in plants and tries) optimize **system-wide health**, while our transport systems often focus on one aspect: costs? Contemporary decision-support systems remain dominated by economic objectives often overlooking system-wide performance. This neglect is reflected in societal and environmental externalities, including traffic congestion, noise pollution, traffic accident risks, GHG emissions, particulate matter, and land misuse.

This no longer meets the demand for creating sustainable and resilient urban logistics systems. Analogous to how Gross Domestic Product serves as a demonstrably inadequate indicator of genuine economic welfare, cost minimization constitutes a fundamentally shallow metric for evaluating system-wide logistics performance.

"Just as evolutionary pressures drive low energy consumption in biological vascular systems; urban logistics requires more sophistication than simply moving products from A to B."

Moreover, current optimization models do not provide a transparent view on the impact of alternative delivery strategies. Consider, for instance, a consumer ordering a book from an online retailer with home-delivery. What would be the total impact compared to alternative delivery strategies, such as utilizing parcel lockers, or other procurement strategies, such as retrieving the book from a local bookstore through personal transport? Can we empower consumers to make better decisions through increased transparency. Or in other words:

"How about a Nutri-Score for urban logistics?"

Room For Improvement

Current optimization and assessment approaches suffer from three shortcomings:

1. **Oversimplified objectives.** Most vehicle routing models (VRP models) reduce fleet composition to capacity units, ignoring differences in environmental and societal impacts.
2. **Subjective weighting.** When societal and environmental externalities are included (for example Green VRPs), externalities are converted to cost parameters, creating models vulnerable to interpretive latitude and analytical misconduct.
3. **Missing transparency.** Consumers cannot easily compare the total impact of their (online shopping) decisions. For example, what is the difference between home delivery, delivery in a parcel locker, or pickup from a local store?

What index can empower people to make more informed decisions about their shopping behaviour and delivery strategies that truly captures the overall impact of urban cargo transport?

Support Your Locals

In terms of 'impact' we take a narrow lens in this work: the neighbourhood level (for example the boundaries of a municipality). Although this lens ignores the (global) upstream journey of the products being transported, and the impact within this lens may appear negligible in the big scheme of things; our own environment matters to people and is something on which we have some say as well as control. For example, we may **'support your locals'** by buying a book at a local bookstore instead of having it home delivered by a large conglomerate.

To empower consumers, focus on the local ecosystem is helpful because we are aware of its presence viscerally and care about its well-being. What's missing is a transparent measure that allows consumers to see the impact of their decisions on their local ecosystem and, subsequently, adjust their decisions accordingly given their beliefs and values.

This is conceptually similar to the Nutri-Score for food products (although not without its flaws) that demonstrates the nutritional value in a single value (A to E) based on a wider set of metrics (for example calories, fat content, and salt levels).

Why not LCA?

One might argue that the method of Life Cycle Assessment (LCA) can be a solution to the challenge presented. However, LCA approaches are typically static and linear and with a coarse spatial and temporal resolution, making it difficult to capture local impacts. Furthermore, LCA is designed for product systems with defined life cycles, whereas logistics is a continuous, service-based activity.

This work presents a first attempt to develop an index that can replace existing metrics to truly represent the overall performance of urban cargo transport.

Theoretically, this index can be applied to a wide variety of logistics flow (e.g., retail, construction, and e-commerce), we scope down our impact assessment of e-commerce deliveries in urban context without fomenting a Luther-like schism. Rather, we aim to reshape how we understand the impacts of transport in urban environments, stimulating meaningful reform.

Nine Fundamental Metrics

Our framework is based on **fundamental** and **non-parametric metrics** that can be classified as *physical* in nature. Our approach is materialized based on two important considerations: (i) to preserve objective quantification with minimal interpretive latitude, and (ii) to reduce the risks of methodological bias or misconduct. These considerations are particularly salient given the frequent bias of organizations to exploit ambiguities in sustainability assessments.

We propose to include the metrics. Note that these are all SI-units. following nine metrics:

Distance [km] Spatial displacement; total distance covered	**Area** [m²] Surface occupation during transit and (un)loading	**Volume** [m³] 3D space occupied; vehicle vs. cargo volume efficiency
Velocity [m/s] Kinematic energy; temporal efficiency of transport	**Mass** [kg] Gravitational burden on infrastructure; wear and tear	**Energy** [J] Power consumption; link with GHG and depletion
Noise [dB(A)] Acoustic pollution; public health and natural well-being	**Heat** [°C] Thermal emissions; urban heat island contribution	**Entropy** [nat] System disorder; fragmentation of deliveries

Figure 1. Nine fundamental metrics capturing the impact of urban transport.

In The Metrics

Distance (D) — Measures the length of the delivery path between origin and destination. Relevant because it cascades into fuel consumption, travel time, and emissions.

Area (A) — Measures the surface footprint occupied by vehicles during transit and (un)loading. Relevant because logistics operations compete with pedestrians, cyclists, and urban green space for limited surface area, directly affecting liveability.

Volume (V) — Measures the 3D space a vehicle claims in the urban environment relative to the cargo it carries. Relevant because it exposes volumetric inefficiency: a delivery van requires vastly more non-cargo space per parcel than a cargo bike.

Velocity (S) — Measures the speed at which goods move through the network. Relevant because both extremes hurt: low speeds increase the temporal footprint on streets, while high speeds amplify energy use, noise, and accident severity.

Mass (M) — Measures the gravitational burden a vehicle places on infrastructure. Relevant because heavier vehicles consume more energy, accelerate road wear, and pose greater safety risks — regardless of whether they carry cargo or run empty.

Energy (E) — Measures the total power consumed to complete a delivery. Relevant because it directly correlates with resource depletion and GHG generation, serving as a proxy for the ecological costs.

Noise (N) — Measures the acoustic pollution imposed on urban soundscapes, in A-weighted decibels. Relevant because elevated noise levels degrade public health and neighbourhood well-being.

Heat (H) — Measures the thermal emissions radiated by vehicles into the urban environment. Relevant because excess heat worsens air quality and intensifies the urban heat island effect, harming both human comfort and urban ecosystems.

Entropy (Σ) — Measures the degree of spatial and temporal fragmentation in a delivery system. Relevant because high entropy — many unique, scattered delivery attempts — drives resource dispersion, while consolidation (e.g., pickup points) reduces system disorder and total impact.

Entropy in Action

Entropy: Consolidation Reduces Disorder

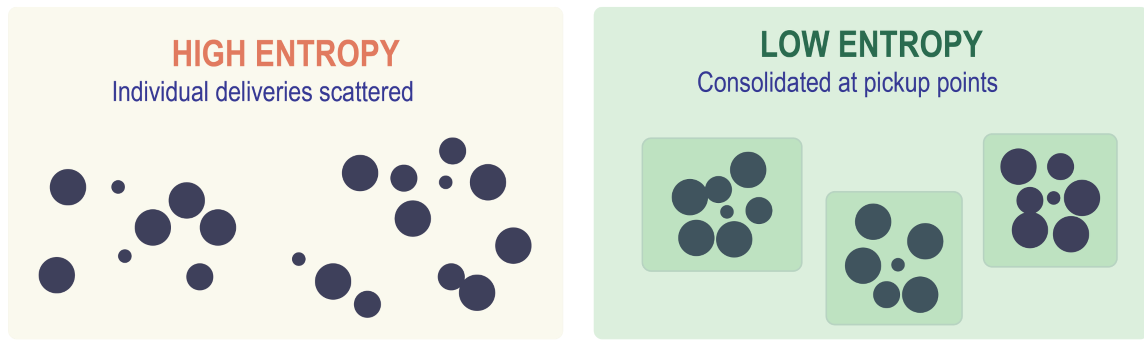

Figure 2. Entropy as a measure of 'disorder' in urban logistics.

Fundamental Impact Factor (FIF)

The metrics serve a dual purpose. First, they allow for the inclusion of novel elements within mathematical optimization models to improve holistic decision-making, although this would inevitably lead to a certain degree of parametrization to balance (conflicting) objectives. Second, and more pertaining to this work, we provide a first attempt to unify the disparate metrics into a single composite measure that facilities transparent, unambiguous, and intuitive comparison between alternative urban logistics schemes. Our Fundamental Impact Factor (FIF) is defined as follows:

$$\mathbf{FIF} = \underbrace{D}_{\text{distance}} \times \underbrace{A}_{\text{area}} \times \underbrace{V}_{\text{volume}} \times \underbrace{S}_{\text{velocity}^{-1}} \times \underbrace{E}_{\text{energy}} \times \underbrace{M}_{\text{mass}} \times \underbrace{H}_{\text{heat}} \times \underbrace{N}_{\text{noise}} \times \underbrace{\Sigma}_{\text{entropy}}$$

For clarity of exposition, normalization constants are omitted and the metrics presented should be interpreted as normalized. It goes beyond the scope of this work to define all normalization constants for, but our proposed general mechanism is to establish theoretical minima as a base for unity, such that all observed values exceed this baseline (i.e., values larger or equal than 1).

Our approach preserves analytical objectivity by avoiding relative importance weights. The multiplicative aggregation ensures that poor performance in one dimension cannot be compensated by excellence in others, thereby maintaining the integrity of each fundamental metric. This methodological stance recognizes that while some degree of transformation is necessary for cross-metric comparison, the critical vulnerability lies not in normalization, but in the subjective value judgments embedded within weighting schemes that reflect organizational priorities rather than physical realities. The FIF thus maintains interpretive clarity by grounding all transformations in objective physical principles rather than in stakeholder preferences, establishing a foundation for a genuine impact factor of urban logistics.

Proxy Measures

The FIF supports granular assessment through metric subsets that serve as proxies for specific impact dimensions:

Proxy	Dimension	Interpretation
M x S	Safety risk	Mass & speed ⟶ accident severity
D x A	Land use	Route length & spatial footprint
V x D	Space claim	3D volume over distance ('cubic kilometers')
H x N	Community impact	Thermal & accoustic pollution

Table 1. Subsets of the FIF serve as proxies for specific impact dimensions.

We are not there yet…

The Fundamental Impact Factor presented here provides a new way of thinking… but we are not there yet. Are you interested in further shaping this work? We would love to hear your opinion. Scan the QR below to reach out, and, who knows, perhaps we can make impact together!

Contact Information

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Scan to reach out!